

Predictive Customer Retention Dashboard using Tableau

Date: 30 April 2026

Dataset Used: Bank Customer Churn Dataset

Dataset Selection:

The dataset was selected because it contains customer demographics, financial details, and a clear target variable (**Exited**) indicating churn. This makes it suitable for predictive modeling and segmentation. It allows calculation of churn probability and identification of high-risk customers for business decision-making.

Objective:

- Predict customer churn using a machine learning model.
- Segment customers into risk levels.
- Identify **Top 100 high-risk customers**.
- Build an interactive dashboard for actionable insights.

Task Performed:

1. Cleaned and prepared the dataset by removing irrelevant columns and converting categorical variables into numerical format.
2. Performed exploratory data analysis to identify patterns such as higher churn among older and inactive customers.
3. Built a Random Forest model to predict churn probability for each customer.
4. Created a new feature for churn probability and segmented customers into Low, Medium, and High risk groups.
5. Prepared the final dataset by combining original data with predicted probability and risk level.
6. Connected the dataset to Tableau and created calculated fields for KPIs and analysis.
7. Designed KPI cards including Total Customers, Churn Rate, High Risk Customers, and Average Probability.

8. Developed visualizations such as Risk Distribution, Churn by Geography, and Age vs Churn Probability.
9. Created a Top 100 Customers table sorted by highest churn probability for targeted retention actions.
10. Applied formatting, filters, and color themes to improve readability and dashboard usability.
11. Finalized and published the dashboard with interactive features for better business insights.

Conclusion:

The dashboard effectively identifies high-risk customers and provides a prioritized list for retention strategies. It converts predictive results into actionable insights, helping businesses reduce churn and improve decision-making.